CONFIDENTIAL. Limited circulation. For review only.

Translating Team Strategy into Closed-Loop Vehicle Control for Multi-Agent Autonomous Racing

Zhishan Wang[1], Kaijie Zhu[2], Zhengyang Liu[3], Luhang Hong[1], Zheng Qing[4], Jacob Junjie Zhang[4], and Rong Su[1]

***Abstract*— Multi-agent autonomous racing requires vehicles to pursue a shared team objective while retaining role-dependent behavior and translating high-level intent into executable control actions. This paper develops a role-aware strategy-to-control framework for this objective-to-execution problem. A common team strategy is combined with vehicle-specific role offsets, while distributed coordination is performed on role-normalized residuals to align shared intent without eliminating functional role differences. The resulting strategy is mapped through an explicit interface to vehicle references, cooperative behavior, and controller parameters. Convergence of the anchored coordination mechanism is established under connectivity, target-pinning, and spectral conditions, together with a tracking bound for time-varying objectives. Controlled simulations validate the strategy-to-execution mapping and the predicted Lead–Support behavior. Results further show that neighbor coordination restores team intent under restricted target access, while successful task execution still depends on downstream planning feasibility. The resulting architecture provides an interpretable and diagnosable connection from team strategy to vehicle control.**

***Index Terms*— Autonomous racing, multi-agent systems, distributed coordination, cooperative control, vehicle control, system autonomy, autonomous vehicles.**

I. INTRODUCTION

Autonomous racing provides a demanding testbed for control, planning, and autonomous decision-making, where vehicles operate under tight dynamic and geometric constraints and close to their performance limits. The challenge becomes deeper in multi-vehicle team racing: vehicles must not only track feasible trajectories, but also execute different local actions according to shared team objectives, teammate roles, and opponent states. The central question is therefore how a global team strategy can be translated into vehicle-level execution, so that high-level preferences such as aggressiveness, cooperation, and risk acceptance systematically influence local references and control behavior.

[1]Zhishan Wang, Luhang Hong, and Rong Su are with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798, Singapore. z.wang103@e.ntu.edu.sg, hong0283@e.ntu.edu.sg, rsu@ntu.edu.sg.

[2]Kaijie Zhu is with the Georgia Institute of Technology, Atlanta, GA 30332, USA. kzhu324@gatech.edu.

[3]Zhengyang Liu is with the Department of Control Science and Engineering, Harbin Institute of Technology, Harbin 150001, People's Republic of China, and is currently a joint Ph.D. student with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. SonnenlichtLiu@outlook.com.

[4]Zheng Qing and Jacob Junjie Zhang are with Purdue University, West Lafayette, IN 47907, USA. qingz@purdue.edu, zhang5933@purdue.edu.

Existing work has addressed autonomous-vehicle and autonomous-racing execution from several complementary directions. Learning-enhanced model predictive control has been developed for racing under model uncertainty [1], while fast iterative MPC has been studied for high-precision nonlinear vehicle tracking [2]. Recent learning-based approaches further consider constrained reinforcement learning for racing safety [3], end-to-end sim-to-real transfer [4], and strategy-conditioned policies that allow a single racing vehicle to adapt among multiple driving objectives [5].

Multi-agent racing introduces an additional interaction and coordination problem. Game-theoretic formulations model competitive multi-vehicle racing through constrained dynamic games [6], while hierarchical two-team approaches assign different tasks and cooperative roles to multiple vehicles [7]. These works address interaction-aware planning and differentiated behavior, but leave a distinct interface problem: how a shared team objective should be distributed under partial information, differentiated by vehicle roles, and translated into executable local control objectives.

This objective-to-control problem presents two key challenges. First, a shared objective should not imply identical local strategies: direct consensus may erase the role differentiation required for team behavior. Second, strategy agreement does not by itself guarantee successful physical execution, since coordinated intent must still be realized through feasible references, tactical decisions, and low-level control. Effective coordination must therefore reconcile team-level agreement with role-dependent local objectives while preserving a realizable path from strategy to execution.

This work extends our earlier studies on autonomous racing control [8] and strategy-driven multi-agent racing [9]. We develop a role-aware strategy-to-control framework in which each team objective is encoded by a common strategy target and vehicle roles are represented through strategy-space offsets. Coordination is performed on the role-normalized residual $\zeta_i = z_i - \varpi_i$, allowing the common component to align while preserving prescribed role differences. An explicit interface then maps the resulting local strategy to vehicle references, cooperative behavior, and controller parameters before closed-loop execution. The layered structure also separates information, coordination, planning, and execution, enabling mechanism-level diagnosis of team behavior.

The main contributions of this work are:

- We formulate a role-aware distributed coordination

Manuscript 192 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received October 7, 2026.

mechanism that aligns a shared team objective while preserving vehicle-specific strategy offsets, together with convergence results for anchored coordination and changing team targets.
- We develop an explicit strategy-to-control interface that maps high-level strategy variables to executable references, cooperation behavior, and controller weights, including an analytical relation between the Support strategy and the resulting Lead–Support spacing.
- We provide controlled simulation evidence that separates the functions of the proposed layers. The experiments validate the strategy-to-execution and role-to-physical mappings, identify the information value of neighbor coordination under restricted target access, and distinguish coordination success from downstream planning feasibility, providing a mechanism-level view of team execution.

II. Problem Formulation

A. Multi-Agent Racing System

Consider a racing team composed of a set of autonomous vehicles $V_T = \{1, \dots, N_T\}$, where N_T denotes the number of team vehicles. External competing vehicles are indexed by the set V_O. The physical control loop is updated with sampling period T_s, while the team-level strategy is updated every $T_c = mT_s$, where $m \in \mathbb{N}_{>0}$ denotes the number of control steps within one strategy-update interval.

For each team vehicle $i \in V_T$, define the Frenet state

$$x_i \triangleq [s_i, d_i, v_i, e_{\psi,i}]^\top \in \mathbb{R}^4 \tag{1}$$

and input

$$u_i \triangleq [\delta_i, a_i]^\top, \tag{2}$$

where s_i, d_i, v_i, and $e_{\psi,i}$ denote the accumulated track progress, lateral offset, speed, and heading error, while δ_i and a_i are the steering angle and longitudinal acceleration, respectively.

Let $\kappa_i \triangleq \kappa(s_i)$ and ℓ denote the local track curvature and vehicle wheelbase, respectively. The nominal Frenet bicycle dynamics are

$$\begin{aligned} \dot{s}_i &= \frac{v_i \cos e_{\psi,i}}{1 - \kappa_i d_i}, & \dot{d}_i &= v_i \sin e_{\psi,i}, \\ \dot{v}_i &= a_i, & \dot{e}_{\psi,i} &= \frac{v_i}{\ell} \tan \delta_i - \kappa_i \dot{s}_i. \end{aligned} \tag{3}$$

The Frenet representation in (3) is considered on track regions satisfying $1 - \kappa(s_i) d_i > 0$, so that the local centerline projection remains well defined. Vehicle states evolve through (3), whereas team-level strategy and coordination determine the references and control objectives to be executed by this physical layer.

B. Team Objective, Roles, and Strategy State

Let $\gamma \in \Gamma$ denote the current team-level racing mode, where Γ is a finite set of admissible objectives, e.g., Conservative, Balanced, Attack, and Protect. Each mode is represented by a common strategy target

$$b_\gamma \triangleq [b_{\alpha,\gamma}, b_{c,\gamma}, b_{\eta,\gamma}]^\top \in [0,1]^3, \tag{4}$$

whose components encode the nominal aggressiveness, cooperation level, and risk acceptance of the team objective, respectively.

Each vehicle $i \in V_T$ is assigned a discrete role $\rho_i \in \mathcal{R}$, where $\mathcal{R}$ denotes the role set (e.g., Lead and Support). Its local strategy state is

$$z_i \triangleq [\alpha_i, c_i, \eta_i]^\top \in [0,1]^3, \tag{5}$$

where α_i, c_i, and η_i denote aggressiveness, cooperation/support coupling, and risk acceptance, respectively.

To encode differentiated vehicle behavior under a shared team objective, let $\varpi_i \triangleq \varpi(\gamma, \rho_i) \in \mathbb{R}^3$ denote the role-dependent strategy offset of vehicle i. The desired vehicle-level strategy is therefore

$$z_i^\star = b_\gamma + \varpi_i. \tag{6}$$

Rather than directly coordinating z_i, we introduce the role-normalized residual

$$\zeta_i \triangleq z_i - \varpi_i, \tag{7}$$

so that coordination toward the common target b_γ preserves the prescribed role difference $z_i^\star - z_j^\star = \varpi_i - \varpi_j$.

C. Information Structure and Problem Statement

The team communication structure is modeled by an undirected weighted graph $\mathcal{G} = (V_T, \mathcal{E}, W)$, where $\mathcal{E} \subseteq V_T \times V_T$ is the communication edge set and $W = [w_{ij}] \in \mathbb{R}^{N_T \times N_T}$ is a symmetric nonnegative weight matrix. The neighbor set of vehicle i is $\mathcal{N}_i \triangleq \{j \in V_T : (i,j) \in \mathcal{E}\}$. The corresponding weighted graph Laplacian is

$$L \triangleq \operatorname{diag}(W\mathbf{1}) - W. \tag{8}$$

Access to the current team target is represented by $d_i^{\mathrm{pin}} \in \{0,1\}$, where $d_i^{\mathrm{pin}} = 1$ indicates that vehicle i directly receives b_γ, and $d_i^{\mathrm{pin}} = 0$ otherwise. Define

$$D \triangleq \operatorname{diag}(d_1^{\mathrm{pin}}, \dots, d_{N_T}^{\mathrm{pin}}). \tag{9}$$

We distinguish direct target access from physical-state observation: a vehicle without access to b_γ may still observe the racing environment and exchange strategy information with its neighbors. Let $\mathcal{I}_i$ denote the locally available information of vehicle i, including its own state, track information, received neighbor strategy packets, and locally available opponent information.

The objective is to design a coordination-and-execution mechanism that drives the role-normalized strategies toward the common target while preserving role differentiation,

$$\zeta_i \to b_\gamma, \qquad z_i \to z_i^\star = b_\gamma + \varpi_i, \tag{10}$$

and maps each resulting strategy to executable vehicle-level control objectives through

$$\theta_i = \Phi(z_i, \rho_i, \mathcal{I}_i), \tag{11}$$

where θ_i collects the local reference and control parameters used by the execution layer. The resulting architecture therefore realizes the chain

$$\gamma \longrightarrow z_i \longrightarrow \theta_i \longrightarrow u_i, \tag{12}$$

Manuscript 192 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received October 7, 2026.

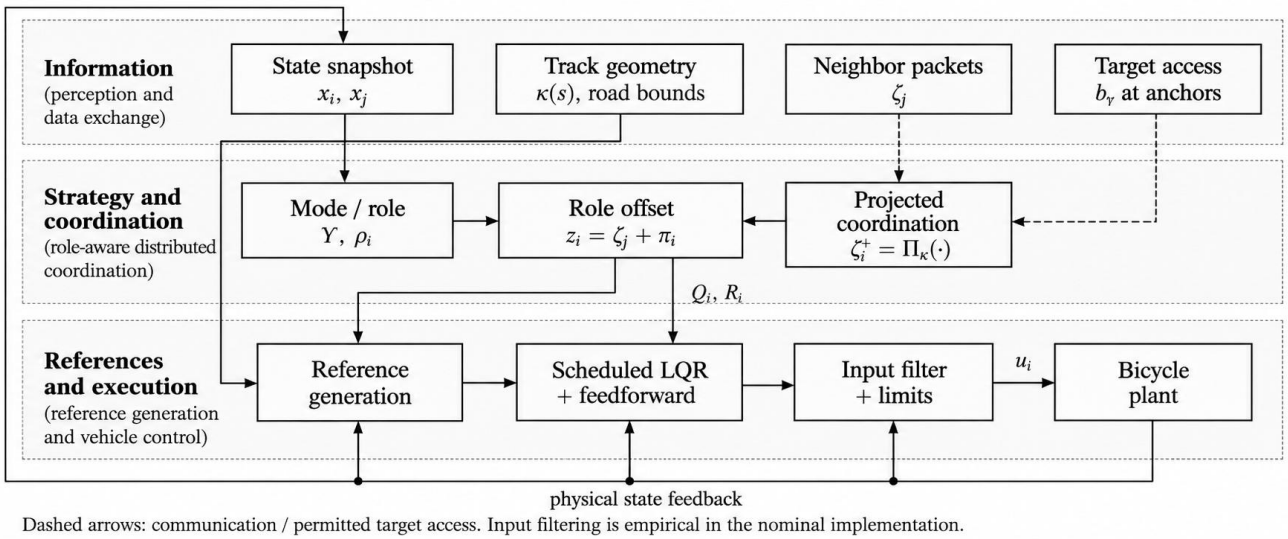


Fig. 1. Layered strategy-to-control architecture. State and track information, neighbor strategy residuals, and authorized team-target access feed the role-aware coordination layer. The resulting local strategy modulates reference generation and controller parameters before closed-loop vehicle execution.

while allowing different team vehicles to execute distinct role-dependent behaviors under the same team objective. Figure 1 summarizes the corresponding closed-loop information, strategy, and execution layers.

For consistency with the derivation, role reconstruction throughout this paper is defined as $z_i = \zeta_i + \varpi_i$. The fixed framework illustration uses the legacy symbol π_i for the role offset; the local residual in this reconstruction is ζ_i.

III. Role-Aware Strategy-to-Control Coordination

A. Role-Normalized Distributed Coordination

A shared team objective should not force identical vehicle strategies. For any pair of team vehicles i and j, we therefore construct the desired strategy difference from the node-wise role offsets as

$$\Delta_{ij}^{\mathrm{role}} \triangleq \varpi_i - \varpi_j. \tag{13}$$

Such a construction is cycle-consistent: the signed sum of $\Delta_{ij}^{\mathrm{role}}$ along every closed cycle is zero. Conversely, on a connected graph, this cycle condition is sufficient for the existence of node offsets realizing the prescribed pairwise differences.

Because $z_i \in [0,1]^3$, define the common feasible residual set

$$\mathcal{K}_\gamma \triangleq \bigcap_{i \in V_T} \left([0,1]^3 - \varpi_i \right), \tag{14}$$

where $[0,1]^3 - \varpi_i$ denotes the set of residuals ζ_i for which $\zeta_i + \varpi_i \in [0,1]^3$. We assume $\mathcal{K}_\gamma$ is nonempty and contains the common target b_γ. The neighbor-coupling term follows the standard consensus structure for networked multi-agent systems [10].

At each strategy update k, vehicle i updates its role-normalized strategy according to

$$\zeta_i[k+1] = \Pi_{\mathcal{K}_\gamma}\Big(\zeta_i[k] + \mu \sum_{j \in \mathcal{N}_i} w_{ij} \big(\zeta_j[k] - \zeta_i[k] \big) + \nu d_i^{\mathrm{pin}} \big(b_\gamma - \zeta_i[k] \big) \Big), \tag{15}$$

followed by

$$z_i[k+1] = \zeta_i[k+1] + \varpi_i, \tag{16}$$

where $\mu > 0$ and $\nu > 0$ are the neighbor-coupling and target-anchor gains, respectively, and $\Pi_{\mathcal{K}_\gamma}(\cdot)$ denotes Euclidean projection onto $\mathcal{K}_\gamma$. The resulting coordination mechanism and its downstream strategy-to-execution interface are summarized in Fig. 2.

In Fig. 2, the fixed illustration uses the legacy labels π_i, K, and $d_{\mathrm{pin},i}$; these correspond to ϖ_i, $\mathcal{K}_\gamma$, and d_i^{pin}, respectively, in the analytical notation. Under the nominal full-state observation used in the present experiments, the available Lead-speed estimate satisfies $\hat{v}_L = v_L$.

Theorem 1 (Role-aware convergence). Let

$$H \triangleq \mu L + \nu D. \tag{17}$$

If $\mathcal{G}$ is connected, at least one vehicle is pinned to the team target, $b_\gamma \in \mathcal{K}_\gamma$, and $\lambda_{\max}(H) < 2$, then $H \succ 0$ and

$$\|\epsilon[k]\|_2 \le q^k \|\epsilon[0]\|_2, \qquad q \triangleq \|I - H\|_2 < 1, \tag{18}$$

where $\epsilon[k] \triangleq \mathrm{col}_{i \in V_T}(\zeta_i[k] - b_\gamma)$. Consequently,

$$z_i[k] \to b_\gamma + \varpi_i, \qquad i \in V_T. \tag{19}$$

Proof sketch: Connectivity together with target pinning gives $H \succ 0$. The spectral condition implies $\|I - H\|_2 < 1$, while Euclidean projection onto the common convex set is nonexpansive. Applying these two properties to (15) yields (18).

Manuscript 192 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received October 7, 2026.

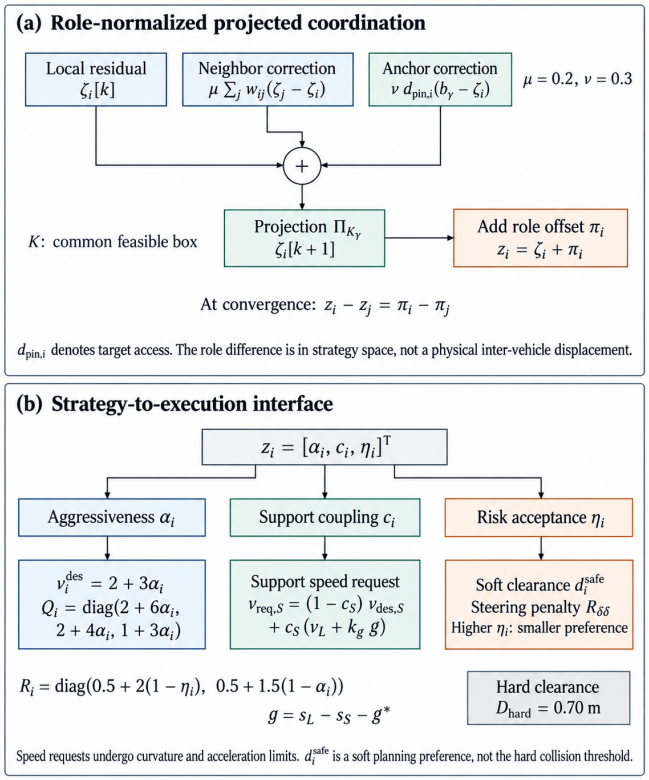


Fig. 2. Core mechanisms of the proposed framework. (a) Role-normalized projected coordination combines the local residual with neighbor and target-anchor corrections before projection and role-dependent strategy reconstruction. (b) The coordinated strategy $z_i = [\alpha_i, c_i, \eta_i]^\top$ modulates desired motion, cooperation behavior, controller weighting, and clearance preference before physical execution.

B. Dynamic Team Objectives

Racing objectives may change as the tactical situation evolves. Let $b[k]$ denote the team target at strategy step k, and let $r[k]$ collect additive coordination perturbations such as bounded message staleness. Define

$$\epsilon[k] \triangleq \text{col}_{i \in V_T} \left(\zeta_i[k] - b[k]\right). \tag{20}$$

Suppose the admissible coordination updates satisfy a uniform contraction factor $q < 1$. Then the role-normalized strategy error obeys

$$\|\epsilon[k+1]\|_2 \le q\|\epsilon[k]\|_2 + \|r[k]\|_2 + \sqrt{N_T}\, \|b[k+1] - b[k]\|_2. \tag{21}$$

Thus, the strategy transient is explicitly driven by both communication perturbations and changes in the team objective.

When a vehicle role changes, we preserve the residual state ζ_i and reconstruct the strategy using the new role offset,

$$\zeta_i^+ = \zeta_i^-, \qquad z_i^+ = \zeta_i^- + \varpi_i^+, \tag{22}$$

while the physical vehicle state x_i remains continuous.

C. Strategy-to-Control Interface

The coordination layer determines the local strategy z_i, but the strategy must still be translated into executable vehicle objectives. The interface illustrated in Fig. 2(b) is defined as

$$\theta_i \triangleq \left\{Q_i,\ R_i,\ v_i^{\text{req}},\ P_i^{\text{ref}},\ d_i^{\text{safe}}\right\}, \tag{23}$$

where Q_i and R_i are the LQR state and input weights, v_i^{req} is the requested speed, P_i^{ref} is the selected reference path, and d_i^{safe} is a soft clearance preference used by the planner.

For $z_i = [\alpha_i, c_i, \eta_i]^\top$, the nominal interface uses

$$\begin{aligned} Q_i &= \text{diag}(2 + 6\alpha_i,\ 2 + 4\alpha_i,\ 1 + 3\alpha_i), \\ R_i &= \text{diag}\left(0.5 + 2(1 - \eta_i),\ 0.5 + 1.5(1 - \alpha_i)\right), \\ v_i^{\text{des}} &= v_{\text{lo}} + (v_{\text{hi}} - v_{\text{lo}})\alpha_i, \end{aligned} \tag{24}$$

where v_{lo} and v_{hi} are the minimum and maximum strategy-level speed requests.

For a Lead–Support pair, define the longitudinal gap error

$$g \triangleq s_L - s_S - g^\star, \tag{25}$$

where $g^\star > 0$ is the desired Lead–Support spacing. The Lead requests $v_L^{\text{req}} = v_L^{\text{des}}$, whereas the Support uses

$$v_S^{\text{req}} = (1 - c_S)v_S^{\text{des}} + c_S\left(\hat{v}_L + k_g g\right), \tag{26}$$

where $c_S \in (0, 1]$ is the Support coupling component of z_S, $\hat{v}_L$ is the locally available Lead-speed estimate, and $k_g > 0$ is the gap-feedback gain.

The requested speed is converted to a dynamically feasible reference. Let $v_i^{\text{cap}}[k]$ denote the speed cap obtained from the previewed path curvature and braking envelope, and define

$$\mathcal{V}_i[k+1] \triangleq [0, v_i^{\text{cap}}[k+1]] \cap [v_i^r[k] + a_{\min}T_s,\ v_i^r[k] + a_{\max}T_s], \tag{27}$$

where $a_{\min} < 0 < a_{\max}$ are the longitudinal acceleration bounds. When $\mathcal{V}_i[k+1]$ is nonempty, the executed speed reference is

$$v_i^r[k+1] = \Pi_{\mathcal{V}_i[k+1]}\left(v_i^{\text{req}}[k+1]\right). \tag{28}$$

Given the feasible reference $y_i^r = [d_i^r, e_{\psi,i}^r, v_i^r]^\top$, define the tracking error $e_i = y_i - y_i^r$, where $y_i = [d_i, e_{\psi,i}, v_i]^\top$, and the scaled error $\chi_i = D_e^{-1}e_i$. The nominal control command is

$$u_i^{\text{nom}} = u_i^{\text{ff}} - D_u K_i \chi_i, \tag{29}$$

where $\mathbf{u}_i^{\text{ff}}$ is the reference feedforward input, D_e and D_u are fixed state- and input-scaling matrices, and K_i is the discrete-time LQR gain obtained from the local frozen model using Q_i and R_i [11]. The interface in Fig. 2(b) is now made explicit.

Proposition 2 (Role-dependent support behavior). Consider straight-track motion with constant $c_S \in (0, 1]$, perfect Lead-speed observation, and exact tracking of the requested speeds. Then (26) yields

$$\dot{g} = -c_S k_g g + (1 - c_S)\left(v_L^{\text{des}} - v_S^{\text{des}}\right), \tag{30}$$

and hence

$$g_\infty = \frac{1 - c_S}{c_S k_g}\left(v_L^{\text{des}} - v_S^{\text{des}}\right). \tag{31}$$

Thus, the cooperation component c_S has a direct and predictable effect on the physical Lead–Support spacing.

Manuscript 192 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received October 7, 2026.

TABLE I

MAIN-TEXT EXPERIMENTS AND ISOLATED MECHANISMS.

Study	Configurations	Isolated question
E1	Fixed (A0), reference-only (A1), weight-only (A2), full interface (A3)	Does the strategy interface produce the intended physical response?
E2	Identical (R0), direct roles (R1), coordinated roles (R2)	Do role-dependent strategies produce predictable physical behavior?
E3	Fixed, Top, Neighbor, Full under I-All and I-Pin	When does neighbor coordination provide new target information?
X1	Full/I-Pin with $H \in \{1,3\}$ s and $\Delta d \in \{0.7, 0.9\}$ m	Does downstream tactical feasibility limit task execution?

IV. EXPERIMENTAL EVALUATION

A. Setup and Evaluation Protocol

The experiments are organized as mechanism-level ablations rather than as a single controller-ranking benchmark. Each study isolates one link in the proposed objective-to-control chain while retaining the same physical plant and execution stack whenever that link is not under test. The main studies and their controlled comparisons are summarized in Table I.

The strategy-to-control pipeline is implemented in MATLAB-compatible code and executed using GNU Octave 10.3.0. The nominal plant is the kinematic bicycle model in (3) with wheelbase $\ell = 0.33$ m. Control and strategy updates use $T_s = 0.02$ s and $T_c = 0.10$ s, respectively, while the continuous plant is integrated by fourth-order Runge–Kutta with a 0.004 s substep. Each condition uses 20 paired held-out test seeds; development uses separate seeds. The main evaluation contains 1180 trajectories, with 100 additional trajectories used for explicitly labeled diagnostic extensions. Reported confidence intervals are obtained from 10,000 bootstrap resamples of complete paired trajectories.

We use three primary physical and strategy-level metrics. For an evaluation horizon T, vehicle progress, maximum strategy-target error, and Lead–Support gap RMS are defined as

$$\begin{aligned} \Delta s_i(T) &\triangleq s_i(T) - s_i(0), \\ E_{\mathrm{tar}}[k] &\triangleq \max_{i \in V_T} \|\zeta_i[k] - b[k]\|_2, \\ \mathrm{RMS}_g &\triangleq \left(\frac{1}{N_s} \sum_{k=0}^{N_s-1} g^2[k] \right)^{1/2}, \end{aligned} \tag{32}$$

where N_s is the number of sampled points in the evaluation window and g is the gap error defined in (25). A pass is declared when the Lead remains more than 1 m ahead of the opponent for at least 2 s without a collision or road-boundary violation.

We use the closed elliptical track S1 for the strategy-interface study, a straight Lead–Support scenario C2 with desired spacing $g^\star = 2.5$ m for the role study, and the multi-vehicle passing scenario C3 for the information and planning studies. All compared methods share the same actuator limits and execution filtering. Across the 1180 main evaluation trajectories, no sampled collision, road-boundary violation, or hard-distance violation was observed.

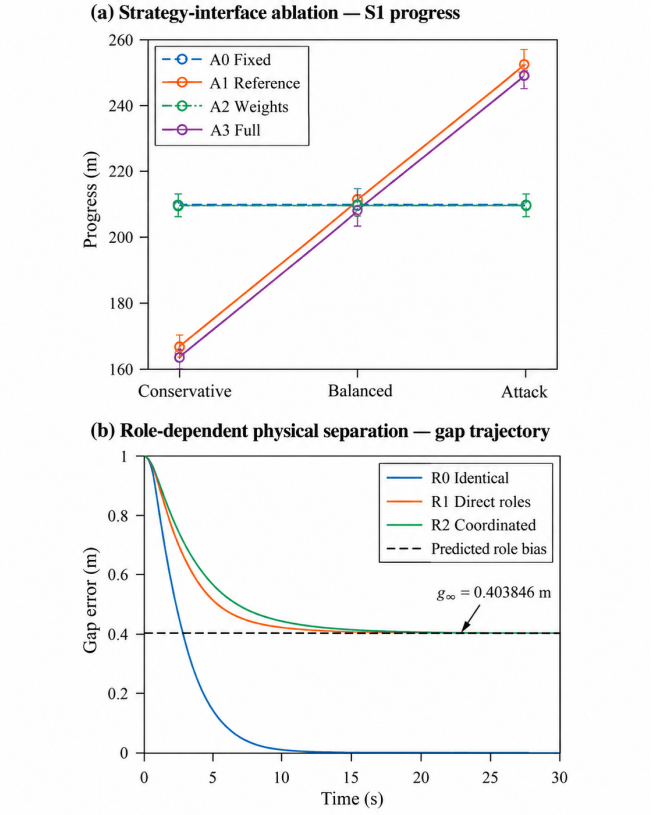


Fig. 3. Strategy and role execution. (a) Strategy-interface ablation on S1: reference-only (A1) and full (A3) modulation changes physical progress with the commanded mode, whereas weight-only modulation (A2) remains close to the fixed baseline (A0). (b) Direct (R1) and coordinated (R2) role strategies converge to the predicted steady-state gap bias $g_\infty = 0.403846$ m.

B. Strategy and Role Execution

We first test whether the strategy variables produce the intended vehicle-level behavior independently of distributed coordination. In E1, the desired strategy is assigned directly, thereby removing the coordination transient. A0 keeps both reference and controller weights at the Balanced setting; A1 modulates only the reference path/speed; A2 modulates only Q_i and R_i; and A3 applies the complete interface.

Figure 3(a) shows the S1 results. The reference-only and full interfaces respond strongly to the commanded mode, whereas the fixed and weight-only configurations remain close to the Balanced progress level. Under Attack, the mean 60-s progress is 209.837 m for A0, 253.079 m for A1, 209.834 m for A2, and 253.076 m for A3. Thus, the high-level strategy is translated into an observable physical response, and the ablation further identifies reference modulation as the dominant pathway for progress in this setting. Weight scheduling alone changes the local tracking response but does not reproduce the progress change when the reference is held fixed.

The role study E2 next examines whether the continuous strategy components have a predictable physical interpretation. In C2, R0 removes the continuous role offset, R1 directly assigns the role-dependent strategy, and R2 generates the same role structure through the distributed coordination

Manuscript 192 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received October 7, 2026.

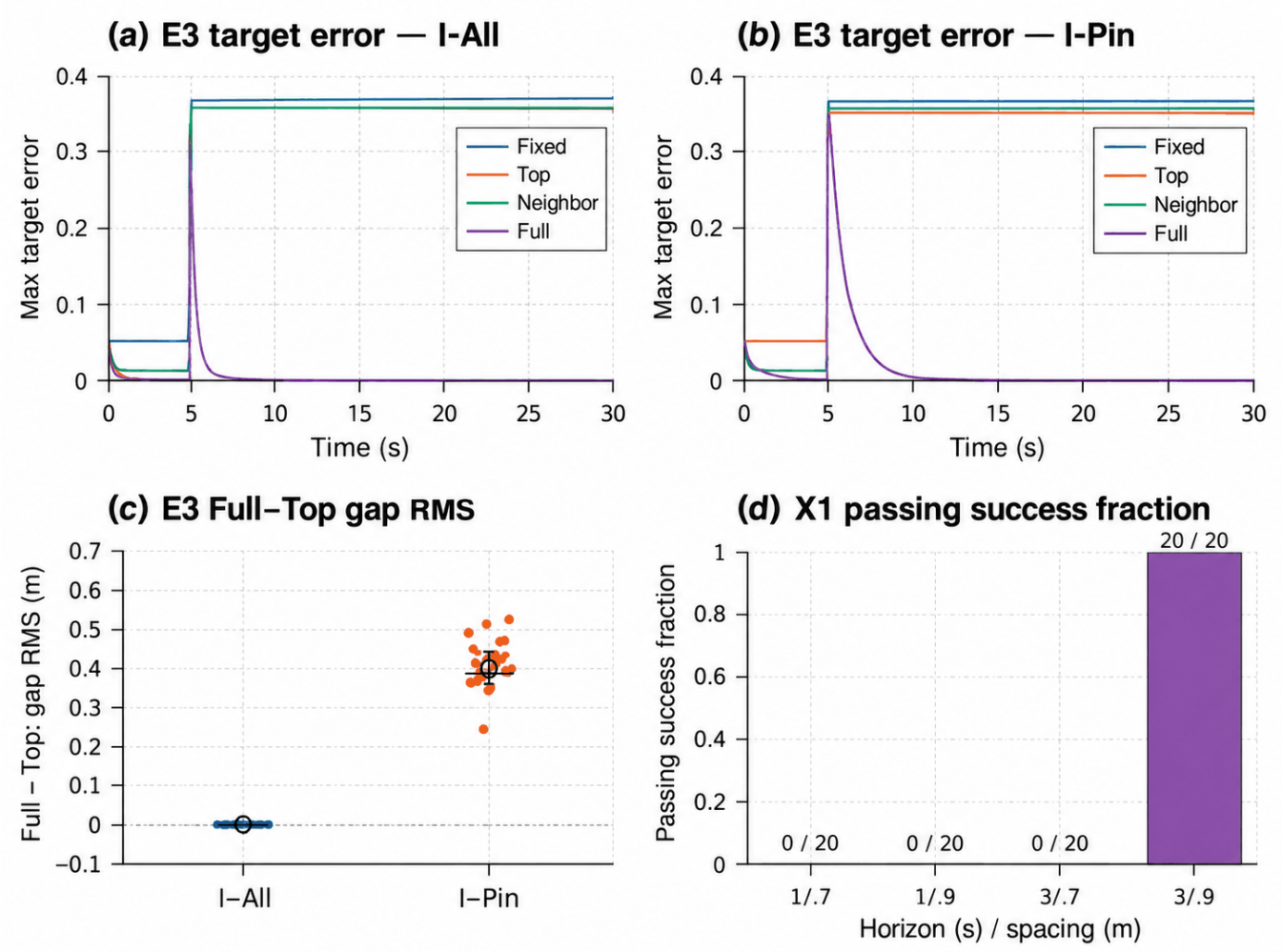


Fig. 4. Information access and planning feasibility. (a)–(b) Neighbor coupling is redundant for target recovery under I-All but propagates the team target under I-Pin. (c) Target recovery does not necessarily reduce physical gap error. (d) With the same Full coordinator, successful passing requires sufficient planning horizon and lane spacing.

mechanism. From (31), the selected parameters $c_S = 0.65$, $k_g = 0.8$, $v_L = 4.55$ m/s, and $v_S^{\mathrm{des}} = 3.95$ m/s predict

$$g_\infty = 0.403846 \text{ m}. \tag{33}$$

As shown in Fig. 3(b), both R1 and R2 converge to 0.403846 m, while R0 converges essentially to zero. The agreement between the analytical prediction and the executed trajectory establishes a direct link from the Support strategy component to measurable Lead–Support behavior. The nearly identical R1 and R2 steady states also show that the distributed mechanism preserves the intended role behavior rather than altering it when the team target is already available.

C. Information Access and Planning Feasibility

We next isolate the information value of neighbor coordination. In E3, Fixed holds the initial strategy, Top uses only the target-anchor term in (15), Neighbor uses only the neighbor term, and Full uses both. Two target-access patterns are considered: I-All uses $D = \mathrm{diag}(1, 1)$, whereas I-Pin uses $D = \mathrm{diag}(1, 0)$ so that only the Lead directly receives the new team target. Physical-state observation is unchanged between the two cases.

In C3, the target changes at $t = 5$ s. Under I-All, Fig. 4(a) shows that both Top and Full recover the new target, since each vehicle already has direct target access. Under I-Pin, however, Top leaves the non-pinned vehicle with a final target error of 0.351421, whereas Full drives the team target error to zero and reaches the prescribed recovery threshold in approximately 2.8 s [Fig. 4(b)]. This comparison isolates the function of the neighbor path: it propagates team intent when direct target access is not globally available.

Improved target consistency, however, does not by itself determine physical task quality. Define

$$\Delta\mathrm{RMS}_g \triangleq \mathrm{RMS}_g^{\mathrm{Full}} - \mathrm{RMS}_g^{\mathrm{Top}}. \tag{34}$$

For C3/I-Pin, Full reduces the final target error to zero, but the paired Lead-progress difference remains below 10^{-8} m and $\Delta\mathrm{RMS}_g$ increases by 0.4075 m (95% CI: $[0.3774, 0.4363]$ m), as shown in Fig. 4(c). Hence, coordination solves the target-information problem, while the physical

outcome remains dependent on the downstream reference and tactical feasible set.

To isolate this downstream limitation, X1 keeps the Full/I-Pin coordinator unchanged and varies only the planner prediction horizon H and candidate-lane spacing Δd. The four settings are $(H, \Delta d) \in \{(1, 0.7), (1, 0.9), (3, 0.7), (3, 0.9)\}$, with units of seconds and meters, respectively. The first three settings produce $0/20$ successful passes, whereas the joint 3 s/0.9 m setting produces $20/20$ successful passes [Fig. 4(d)]. Because the coordinator is identical across all four cases, the change identifies tactical feasibility—rather than additional strategy agreement—as the limiting factor. Coordinated intent therefore becomes successful physical behavior only when the planning layer provides a feasible action set.

Taken together, the experiments validate distinct functions of the proposed architecture. The strategy interface converts abstract team preferences into executable vehicle behavior; role offsets produce predictable physical differentiation; neighbor coordination restores team intent when target access is partial; and the planning study shows that information consistency alone is insufficient without a feasible downstream action set. The layered formulation therefore provides not only coordinated behavior but also a mechanism-level separation of where information, strategy, planning, and execution affect the final racing outcome.

V. Discussion and Conclusion

This paper developed a role-aware strategy-to-control framework for multi-agent autonomous racing. By coordinating the role-normalized strategy residual $\zeta_i = z_i - \varpi_i$, the framework aligns a shared team objective while preserving vehicle-specific role differences, and maps the resulting strategy explicitly to executable references and controller parameters.

The experiments validate the architecture's main mechanisms. The strategy interface produces distinct physical responses under a shared execution structure, and the predicted Lead–Support gap is reproduced by the closed-loop trajectories. Neighbor coordination is particularly useful when target access is restricted, where it propagates the team intent to non-pinned vehicles. At the same time, target agreement alone is not sufficient for task success; coordinated intent must also be supported by a feasible downstream tactical action set.

An important system-level benefit of the layered formulation is mechanism-level diagnosability. Because information access, coordination, planning, and execution are represented separately, task failure can be localized to a specific stage rather than treated as a single end-to-end performance loss. This distinction is also relevant beyond racing. Platooning, cooperative merging, convoy operation, and coordinated lane changing all require vehicles to share a common objective while executing different local roles. The proposed decomposition therefore provides a useful abstraction for connecting cooperative decisions to local vehicle controllers while retaining interpretable failure modes.

Overall, the main contribution of this work is an interpretable closed-loop mechanism connecting shared team intent, role-dependent strategy, distributed information exchange, and vehicle-level control execution. The theoretical analysis establishes role-preserving coordination and its convergence behavior, while the experiments validate the strategy-to-execution mapping, the predicted physical effect of role-dependent parameters, and the information-recovery role of neighbor coordination under restricted target access. The planning diagnostic further shows that successful team behavior depends not only on strategy agreement, but also on the feasibility of downstream tactical actions.

Future work will extend the framework to certified safety filtering, richer vehicle dynamics, communication imperfections, dynamic role assignment, and tighter coordination–planning integration. The framework will also be extended to larger vehicle teams and more general communication topologies, followed by hardware-in-the-loop and physical multi-vehicle validation in more realistic autonomous driving environments.

References

[1] J. Kabzan, L. Hewing, A. Liniger, and M. N. Zeilinger, "Learning-based model predictive control for autonomous racing," *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 3363–3370, 2019, doi: 10.1109/LRA.2019.2926677.

[2] D. Meng, H. Chu, M. Tian, B. Gao, and H. Chen, "Real-time high-precision nonlinear tracking control of autonomous vehicles using fast iterative model predictive control," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 2, pp. 3644–3657, 2024, doi: 10.1109/TIV.2024.3352171.

[3] K. Yu, M. Fu, T. Zhang, and Y. Yang, "Enhancing safety in autonomous racing with constrained reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 6448–6455, 2025, doi: 10.1109/LRA.2025.3566591.

[4] C. Budai, T. Szeles, B. Nemeth, and P. Gaspar, "End-to-end reinforcement learning for autonomous racing: Bridging the sim-to-real gap," in *Proc. American Control Conference (ACC)*, 2025, pp. 200–205, doi: 10.23919/ACC63710.2025.11107738.

[5] Y. Tawfeek and A. Hamoud, "Multi-strategy end-to-end autonomous racing using soft actor-critic," in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2026, pp. 1626–1633, doi: 10.1109/IV66570.2026.11623852.

[6] Y. Jia, M. Bhatt, and N. Mehr, "RAPID: Autonomous multi-agent racing using constrained potential dynamic games," in *Proc. European Control Conference (ECC)*, 2023, pp. 1–8, doi: 10.23919/ECC57647.2023.10178387.

[7] Z. Hu, X. Li, M. Meng, and S. Zhao, "A game theoretic method for two-team multi-player autonomous racing," *IEEE Robotics and Automation Letters*, vol. 9, no. 9, pp. 7581–7588, 2024, doi: 10.1109/LRA.2024.3428128.

[8] K. Zhu, Z. Wang, J. Forrest, K. Desireddi, B. Lee, and S. Chen, "Controller design for autonomous racing vehicle," Zenodo, Apr. 2026, doi: 10.5281/zenodo.19905203.

[9] Z. Wang, K. Zhu, Z. Qing, K. V. Desireddi, J. Zhang, and B. S. Lee, "A strategy-driven multi-agent control framework for autonomous racing," Zenodo, 2026, doi: 10.5281/zenodo.19992669.

[10] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, 2007.

[11] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*. Mineola, NY, USA: Dover Publications, 2007.

Manuscript 192 submitted to 2026 14th International Conference on Systems and Control (ICSC). Received October 7, 2026.